Modeling Crash Injury Severity in New York City: A Bootstrap and Monte Carlo Inference Study

ST 453 Final Project | April 28, 2026

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1. Data Description

The data set analyzed in this project is the New York City Motor Vehicle Collisions – Crashes data set, a publicly available collection of police-reported traffic crashes maintained by the NYC Department of Transportation. This is public data freely available from NYC OpenData at:

https://data.cityofnewyork.us/Public-Safety/Motor-Vehicle-Collisions-Crashes/h9gi-nx95/about_data

The full dataset contains 2,247,203 collisions (obs) and 29 variables. Variables include crash time, borough, injury and fatality counts by road-user type (pedestrians, cyclists, motorists), up to five contributing factors, and up to five vehicle type codes. The primary response variable is NUMBER.OF.PERSONS.INJURED, a non-negative integer recording total persons injured per crash. This data is updated daily, automatically.

To construct an analysis model, six variables were chosen and retained: CRASH.TIME, BOROUGH, NUMBER.OF.PERSONS.INJURED, CONTRIBUTING.FACTOR.VEHICLE.1, VEHICLE.TYPE.CODE.1, and VEHICLE.TYPE.CODE.2. Rows with missing or blank values in any used field were removed, and a random sample of 20,000 crashes was drawn using set.seed(1). After constructing the regression design matrix and removing incomplete rows, the final analysis sample contains 19,999 observations and 15 model columns (including the intercept). Table 1 summarizes data origin. This data set is well suited for studying crash severity because it records both an injury outcome and rich crash-level contextual information whose practical relevance

extends to transportation safety policy and prevention.

Statistic	Value
Minimum	0
1st Quartile	0
Median	0
Mean	0.3069
3rd Quartile	0
Maximum	15
Std. Deviation	0.6719
% Crashes with 0 Injuries	~78.6%

Predictor	Count	Proportion	Pred	Count	Prop
Night (9PM–5AM)	3,860	19.3%	**Following Too Closely**	730	3.7%
Multi-Vehicle	15,873	79.4%	**Failure to Yield**	1,233	6.2%
Brooklyn	6,474	32.4%	**Unspecified Factor**	7,422	37.1%
Manhattan	4,340	21.7%	**Pickup**	417	2.1%
Queens	5,398	27.0%	**Motorcycle**	176	0.9%
Bronx	2,964	14.8%	**Sedan**	5,865	29.3%
Staten Island (reference)	923	4.6%	**SUV**	6,115	30.6%
Inattention/Distraction	**3,932**	**19.7%**			

Table 1. Summary Statistics for NUMBER.OF.PERSONS.INJURED (n = 19,999)

2. Variables of Interest and Relationships Being Studied

The primary outcome is the number of persons injured per crash, used as a continuous measure of crash severity. The inferential goal is to estimate how the conditional mean varies with observable crash characteristics. The Night indicator (1 = crash between 9 PM and 5 AM) reflects reduced visibility and driver fatigue. MultiVehicle (1 = second vehicle type recorded) captures multi-vehicle involvement. Borough indicators for Brooklyn, Manhattan, Queens, and Bronx (reference: Staten Island) account for geographic differences in traffic density and infrastructure.

Contributing factor indicators cover Inattention (Driver Inattention/Distraction), FollowingTooClosely, FailureToYield (Failure to Yield Right-of-Way), and UnspecifiedFactor. Vehicle-type indicators cover Sedan, SUV, Pickup, and Motorcycle. The analysis focuses on how crash characteristics such as time of day, contributing factors, and vehicle types are associated with expected injury counts, both individually and after controlling for other variables.

3. Statistical Model and Bootstrap Framework

Let Y_i denote the number of persons injured in crash i, and let $X_i \in \mathbb{R}^{15}$ denote the corresponding covariate vector (including the intercept). The relationship between crash

characteristics and injury counts is modeled using a linear regression framework:

$$Y_i = x_i^T \beta + U_i, \quad i = 1, \dots, n,$$

where $\beta \in \mathbb{R}^{15}$ is the vector of unknown regression coefficients and U_i are mean-zero error terms. The primary quantity of interest is the conditional mean function $\mathbb{E}(Y_i|X_i) = X_i^T \beta$, which captures how expected injury counts vary with observable crash characteristics such as nighttime driving, multi-vehicle involvement, and contributing factors. While the linear model provides an interpretable approximation, the outcome variable is a zero-inflated count, so the Gaussian error assumption is unlikely to hold exactly. For this reason, inference is not based solely on parametric assumptions, but instead relies on a bootstrap approach. The design matrix X is treated as fixed, and variability is introduced by resampling the residuals from the fitted model.

4. Bootstrap Data Generation

To estimate the sampling distribution of the regression coefficients without relying on distributional assumptions, a residual bootstrap procedure is used. First, the model is fit to the observed data using ordinary least squares (OLS), yielding estimates $\hat{\beta}$, fitted values $\hat{Y} = X\hat{\beta}$, and residuals $\widehat{u_i} = Y_i - \widehat{Y_i}$. To account for heteroscedasticity induced by leverage, standardized residuals are constructed as:

$$r_i = \frac{\widehat{u_i}}{\sqrt{1-h_i}},$$

where h_i are the diagonal elements of the projection matrix $H = X(X^T X)^{-1} X^T$.

Bootstrap samples are then generated as follows: 1. Sample residuals r_i^* with replacement from the centered standardized residuals. 2. Construct new responses: $Y_i^* =$

$\widehat{Y}_i + r_i^*$. 3. Refit the regression model to obtain bootstrap estimates $\hat{\beta}^{*(b)}$. 4. Repeat this procedure B times (with B=10,000).

The resulting bootstrap distribution $\{\hat{\beta}^{*(1)}, \dots, \hat{\beta}^{*(B)}\}$ is used to estimate standard errors, confidence intervals, hypothesis tests. This approach avoids reliance on Gaussian error assumptions and provides inference directly from the empirical distribution of the data.

5. Computational Approach to Fitting the Model

Coefficients are estimated by ordinary least squares (OLS): $\hat{\beta} = \arg\min_{\beta \in \mathbb{R}^p} ||Y - X\beta||_2^2$. In practice this uses QR decomposition via qr.solve(X,Y) in R, which provides a numerically stable solution without explicitly forming $(X^T X)^{-1}$. For the real data, the design matrix X and response vector Y are constructed, after which the coefficient estimate $\hat{\beta}$ is obtained. Fitted values $\hat{Y} = X\hat{\beta}$, residuals $\widehat{U} = Y - \hat{Y}$, and residual standard deviation $\hat{\sigma} = 0.6719$ are used as the noise scale in the bootstrap procedure. Standard errors use the diagonal of $(X^T X)^{-1}\hat{\sigma}_{unb}^2$where computed via solve(t(X)%*%X).

6. Bootstrap Study

A bootstrap study was conducted to assess the sampling distribution of the OLS estimator without relying on parametric assumptions. Rather than generating synthetic data from a Gaussian model as in the midterm, the bootstrap resamples directly from the empirical distribution of the residuals. Each of B=10,000 replications constructs a new response vector by combining fitted values with resampled residuals, after which the model is refit using the same design matrix X.

The estimated coefficients from each bootstrap sample are recorded, and the resulting sampling distributions are summarized using plots and tables.

This approach preserves the exact covariate structure, sample size (n=19,999), and empirical error behavior observed in the real data. A total of B = 10,000 bootstrap samples were used to ensure stable estimation of sampling distributions and confidence interval quantiles. At this scale, Monte Carlo variability in estimated standard errors and percentile bounds is negligible, and further increases in B would produce only minimal improvements relative to computational cost.

Figures 6–8 display the bootstrap sampling distributions for selected coefficients of interest. Figure 6 shows the bootstrap distribution of the Night coefficient, which is centered near the real-data estimate of 0.08195 and exhibits an approximately symmetric, bell-shaped form. The

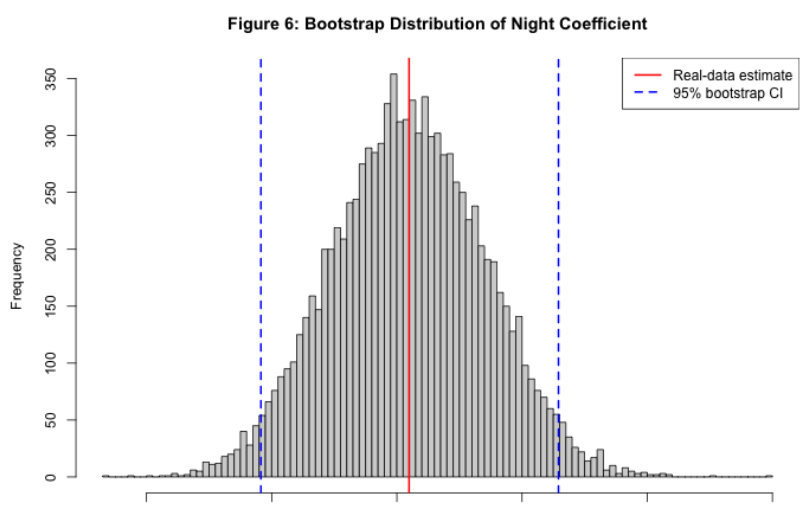

Figure 6. The red line indicates the real-data estimate, and the dashed blue lines represent the 95% bootstrap confidence interval.

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Figure 7 shows the bootstrap distribution for the MultiVehicle coefficient, and Figure 8 shows the corresponding distribution for the FailureToYield coefficient. These distributions exhibit similar behavior, with symmetric shapes centered at their respective real-data estimates and no evidence of heavy skewness or extreme outliers. Across all three coefficients, the spread of the bootstrap distributions aligns closely with the reported bootstrap standard errors, indicating consistent estimation variability.

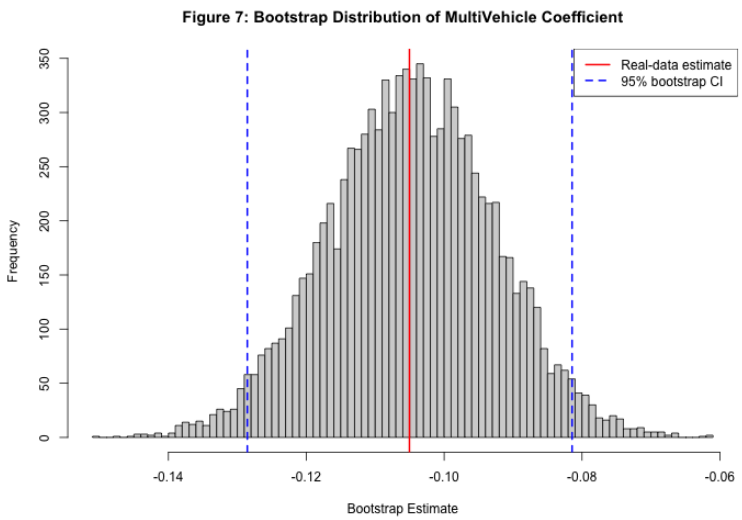

Figure 7. The red line indicates the real-data estimate, and the dashed blue lines represent the 95% bootstrap confidence interval.

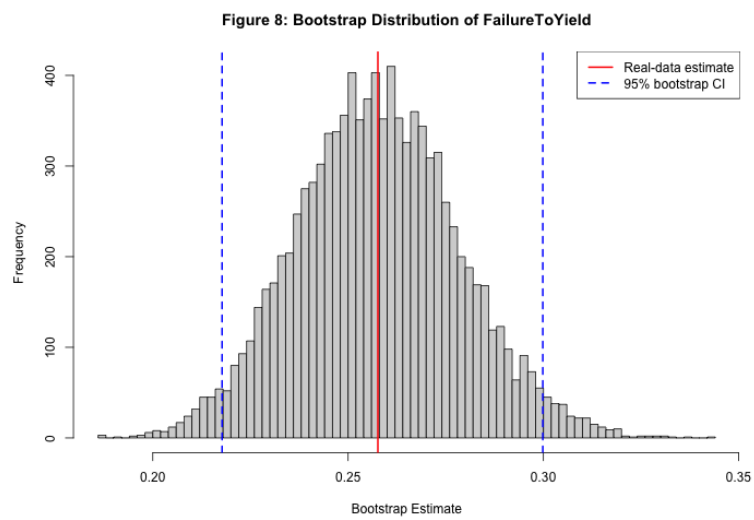

Figure 8. The red line indicates the real-data estimate, and the dashed blue lines represent the 95% bootstrap confidence interval.

Table 4 summarizes the bootstrap inference results across all model coefficients, including bootstrap standard errors, confidence intervals, and p-values. The table confirms the patterns observed in Figures 6–8, showing that the bootstrap estimates are stable and that the empirical sampling distributions provide reliable measures of uncertainty. Together, Figures 6–8 and Table 4 demonstrate that the OLS estimator produces well-behaved sampling distributions under the bootstrap procedure, supporting its use for inference in this setting.

Parameter	Estimate	Parametric SE	Bootstrap SE	Parametric CI Lower	Parametric CI Upper	Bootstrap_CI_Lower	Bootstrap_CI_Upper
Intercept	0.366793	0.027793	0.027871	0.312316	0.42127	0.313189	0.421973
Night	0.081951	0.012073	0.012214	0.058287	0.105614	0.05828	0.105823
MultiVehicle	-0.105036	0.012061	0.012069	-0.128677	-0.081395	-0.128554	-0.081442
Brooklyn	0.032722	0.024883	0.024819	-0.01605	0.081495	-0.017334	0.079992
Manhattan	-0.083101	0.025637	0.025366	-0.133352	-0.032851	-0.134499	-0.034933
Queens	-0.026604	0.025166	0.024982	-0.075933	0.022724	-0.077508	0.020364
Bronx	0.004188	0.02649	0.026521	-0.047735	0.05611	-0.049455	0.05489
Inattention	0.05497	0.013534	0.013568	0.028441	0.081499	0.028549	0.08179
FollowingClose	0.077749	0.026336	0.026317	0.026129	0.129369	0.027778	0.131293
FailureToYield	0.257621	0.020925	0.020773	0.216606	0.298636	0.21773	0.299842
Unspecified	-0.034781	0.011559	0.011574	-0.057437	-0.012125	-0.057292	-0.012093
Sedan	0.011653	0.012247	0.012318	-0.012352	0.035657	-0.012922	0.035704
SUV	8.9E-05	0.01188	0.011834	-0.023197	0.023375	-0.023532	0.022991
Pickup	-0.060353	0.033921	0.033927	-0.126841	0.006136	-0.123694	0.007975
Motorcycle	0.325619	0.051453	0.051181	0.224766	0.426471	0.231329	0.431246

Table 4. Estimates, bootstrap standard errors, 95% bootstrap confidence intervals vs. parametric for each coefficient.

7. Bootstrap Hypothesis Testing and Confidence Intervals

Inference was conducted for the Night coefficient using both parametric and bootstrap methods. Let β_{night} denote the population coefficient. The hypothesis test is $H_0: \beta_{Night} = 0 \; vs. H_A \neq 0$. The estimated coefficient is $\hat{\beta}_{Night} = 0.08195054$. The parametric standard error is 0.01207275, while the bootstrap standard error is 0.01221351, indicating close agreement between the two methods.

The parametric 95% confidence interval is (0.05828696, 0.10561413), and the bootstrap confidence interval is (0.05828007, 0.10582294). These intervals are nearly identical, and both exclude zero. The parametric p-value is effectively zero, and the bootstrap p-value is also approximately zero, leading to rejection of the null hypothesis in both cases. There is strong evidence that nighttime crashes are associated with a nonzero increase in expected injury counts.

The agreement between bootstrap and parametric inference suggests that the Gaussian approximation used in the midterm is reasonable in this setting. However, the bootstrap provides confirmation of this result without relying on distributional assumptions. Table 4 presents the full

bootstrap inference summary, including standard errors, confidence intervals, and p-values for all coefficients.

8. Fitted Model and Estimates on the Real Data

Using the full analysis sample, the estimated coefficient vector remains $\hat{\beta} = (0.3668, 0.0820, -0.1050, 0.0327, -0.0831, 0.0266, 0.0042, 0.0550, 0.0777, 0.2576, -0.0348, 0.0117, 0.0001, -0.0604, 0.3256)$, with residual standard deviation $\hat{\sigma} = 0.6719, R^2 = 0.0238, and\ adj. R^2 = 0.0231$. These values are unchanged from the midterm analysis, as the bootstrap procedure does not alter the fitted model but instead provides an alternative method for inference.

Several estimates remain noteworthy. The Night coefficient (+0.0820) indicates that nighttime crashes are associated with approximately 0.082 additional injured persons on average, holding other variables constant. The FailureToYield coefficient (+0.2576) represents one of the largest positive effects, reflecting increased severity in right-of-way violations. The Motorcycle coefficient (+0.3256) is the largest positive estimate, consistent with the increased vulnerability of motorcycle riders.

The MultiVehicle coefficient (−0.1050) is negative after controlling for other predictors. This does not imply that multi-vehicle crashes are inherently safer, but rather reflects the conditional relationship once other variables are held fixed. As in the midterm, Figure 5 shows observed versus predicted values, with fitted values concentrated between approximately 0.2 and 0.8 while observed values range from 0 to 15. The low R^2 is expected given the absence of key variables such as speed, weather, and driver impairment.

9. Comparison of Bootstrap and Parametric Inference

The bootstrap results closely mirror the parametric results obtained in the midterm project. Standard errors for all coefficients are nearly identical, and confidence intervals overlap almost exactly. Hypothesis tests lead to the same conclusions across both approaches, with significant variables remaining significant and non-significant variables remaining unchanged.

This consistency indicates that the parametric assumptions used in the midterm, particularly the Gaussian error assumption, do not substantially distort inference in this setting. The bootstrap serves as a robustness check, confirming that conclusions drawn from parametric methods are reliable.

For example, the Night coefficient has a parametric standard error of 0.01207 and a bootstrap standard error of 0.01221, with corresponding confidence intervals (0.05829, 0.10561) and (0.05828, 0.10582), respectively. These values are nearly indistinguishable, providing concrete evidence that parametric and bootstrap inference lead to the same conclusions in this setting.

10. Broader Implications

The fitted model suggests that crash injury severity in New York City is meaningfully associated with several observable characteristics. The positive effects for Night, FailureToYield, and Motorcycle align with established findings in traffic safety research. Reduced visibility, risky intersection behavior, and limited vehicle protection are all associated with increased injury severity.

Nighttime driving may warrant increased attention from a public safety perspective, particularly in the context of driver fatigue and reduced visibility. The strong association with

failure-to-yield crashes highlights the importance of intersection safety and driver compliance with traffic rules. The motorcycle finding reinforces the well-documented risks associated with motorcycle crashes, which often result in more severe injuries due to limited physical protection.

Although the model is observational and does not establish causality, it provides a useful representation of the conditional mean relationship between crash characteristics and injury severity. The consistency of results across both parametric and bootstrap methods further supports the reliability of these findings.

11. Limitations

The response variable is a zero-inflated, right-skewed count, for which a Poisson or negative binomial model may be more appropriate. The linear model is used as an approximation to the conditional mean, prioritizing interpretability over distributional accuracy.

Second, many important variables are not included in the data set. Factors such as vehicle speed, impact angle, weather conditions, driver impairment, and seatbelt use are likely to influence injury severity and may be correlated with the included predictors. The omission of these variables may introduce bias into the estimated coefficients.

Third, the model explains only a small portion of the variation in injury counts, as indicated by $R^2 = 0.0238$. While this is not uncommon for observational data of this type, it highlights the complexity of crash severity and the influence of unobserved factors. Finally, while the bootstrap provides a nonparametric method for inference, it still relies on the assumption that the residuals adequately represent the underlying error structure, which may not hold in all cases.

Data citation: NYC OpenData. *Motor Vehicle Collisions – Crashes*. New York City Department of Transportation. Available at:

https://data.cityofnewyork.us/Public-Safety/Motor-Vehicle-Collisions-Crashes/h9gi-nx95 (accessed March 2026).